

Manasvi Khandelwal

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EDUCATION

Bachelor of Science in Computer Science and Mathematics | *University of Wisconsin–Madison* *Madison, WI*
Expected May 2027

- **GPA:** 4.00
- **Honors:** Dean's List – Fall 2023, Spring 2024, Fall 2024, Spring 2025

PROFESSIONAL EXPERIENCE

Software Engineering Intern | *GE HealthCare* *Waukesha, WI*
Summer 2026

- Developed a **proof-of-concept** for adding persistence to **76 microservices** in the MR (Magnetic Resonance) imaging system, using **Java Spring Boot** and **PostgreSQL** to replace in-memory state with durable storage.
- Designed schema and data access layers to integrate persistence into the existing microservice architecture with minimal disruption to service contracts and runtime behavior.
- Worked with internal **Artifactory** repositories and existing **CI/CD pipelines** to manage dependencies and validate automated builds, ensuring a smooth rollout across the microservice fleet.

ML Research Intern | *UW-Madison Math Department* *Madison, WI*
Jan 2025 – Aug 2025

- Conducted research with large, noisy datasets, to train **neural networks** to accelerate numerical solvers for Partial Differential Equations (PDEs).
- Developed and trained **feedforward networks**, **CNNs**, and **ResNets** in **PyTorch** on **CUDA**-enabled GPUs using spectral element data from **MATLAB**, optimizing polynomial degrees and hyperparameters for improved accuracy and stability.
- Integrated models via **ONNX** into PDE solvers, reducing solution error from 10^{-3} to 10^{-4} without increasing computational cost.

Student IT Intern | *College of Agricultural and Life Sciences IT* *Madison, WI*
Sep 2025 – Present

- Provided IT support for **40+ research labs**, diagnosing hardware, networking, and OS-level issues across Linux, Mac, and Windows systems.
- Handled support requests through a **ticketing system** and assisted with **asset management**, including tracking and maintaining lab hardware inventories.

Undergraduate Teaching Assistant | *Department of Computer Science & UW–Math Learning Center* *Madison, WI*
Sep 2024 – Present

- Supported **200+ students** in **Introduction to Computer Systems**.

PROJECTS

CampusLedger | *Python, FastAPI, PostgreSQL, React, Plaid API*

- Built a full-stack college spending tracker ingesting and normalizing transaction data across student-specific categories (dining, textbooks, rent, subscriptions) via the **Plaid API** with a **FastAPI** REST backend.
- Trained a **scikit-learn** categorization model achieving **90%+** accuracy across 8 spending categories, and designed an interactive **React** dashboard visualizing spending trends and predictive budget insights — containerized with **Docker** for consistent deployment.

SoundScape | *React Native (Expo), OpenAI API, Spotify API, Mapbox*

- Built and shipped an **AI-powered mobile navigation app** that dynamically generates personalized **Spotify playlists** by integrating **OpenAI**, real-time weather, and traffic data.
- Designed a **voice-based mood detection** pipeline and architected a multi-API integration layer using **Mapbox**, **Google Places**, and **OpenWeather**, with a demo mode simulating real-time navigation for **MVP validation**.

INVOLVEMENT

Secretary | *DotData – Data Science Club* *University of Wisconsin–Madison*
May 2025 – Present

- Manage weekly newsletter for **600+ members** and coordinate workshops, guest talks, and club events as the primary club contact.
- Oversee finances, website, and exec meetings; organize **MadData hackathon** with **300+ participants**.

Mentoring Chair | *WACM – Women in Computing* *University of Wisconsin–Madison*
Sep 2025 – Present

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Spring Boot, SQL, JavaScript, R, HTML, Flask, MATLAB

Data Science & ML: NumPy, Pandas, Scikit-learn, PyTorch, ONNX, PostgreSQL

Infrastructure & Systems: Linux/Unix, Docker, Git, .NET, Azure, FastAPI, React